Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Core Pure 1 – Series

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Find $\sum_{r=1}^{n} r$. **[2]**

2. Find $\sum_{r=1}^{n} r^2$. **[2]**

3. Find $\sum_{r=1}^{n} r^3$. **[2]**

4. Evaluate $\sum_{r=1}^{10} r$. **[2]**

5. Evaluate $\sum_{r=1}^{5} r^2$. **[2]**

6. Find $\sum_{r=1}^{n} (2r+1)$. **[3]**

7. Find $\sum_{r=1}^{n} (r^2 - r)$. **[3]**

8. Use the method of differences to find $\sum_{r=1}^{n} \left(\frac{1}{r} - \frac{1}{r+1}\right)$. **[4]**

9. Find $\sum_{r=1}^{n} \frac{1}{r(r+1)}$ by partial fractions and telescoping. **[4]**

10. Find $\sum_{r=1}^{n} r(r+1)$. **[3]**

11. Express $\sum_{r=1}^{n} (3r^2 - 2r + 1)$ in terms of n. **[4]**

12. Find $\sum_{r=1}^{20}(2r-1)$. **[3]**

13. Given $\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$, find $\sum_{r=1}^{n}(r+3)$. **[3]**

14. Find $\sum_{r=1}^{n} 2^r$. **[3]**

15. Find the sum of the first n odd numbers. **[2]**

Section B – Hard (Q16–25)

16. Use method of differences to find $\sum_{r=1}^{n} \frac{1}{r(r+2)}$. **[5]**

17. Find $\sum_{r=1}^{n} r^2(r+1)$. **[5]**

18. Show that $\sum_{r=1}^{n} r(r+1)(r+2) = \frac{n(n+1)(n+2)(n+3)}{4}$. **[5]**

19. Find $\sum_{r=1}^{n}\left(\frac{1}{r} - \frac{2}{r+1} + \frac{1}{r+2}\right)$. **[5]**

20. Evaluate $\sum_{r=1}^{100} r^3 - \left(\sum_{r=1}^{100} r\right)^2$ and comment. **[4]**

21. Find a closed form for $\sum_{r=1}^{n}(2r-1)^2$. **[4]**

22. Use induction or formulae to find $\sum_{r=1}^{n} r(3r-1)$. **[4]**

23. Find $\sum_{r=1}^{n} \frac{r}{2^r}$ (hint: arithmetico-geometric series). **[6]**

24. Given $u_r = \frac{1}{(2r-1)(2r+1)}$, find $\sum_{r=1}^{n} u_r$. **[5]**

25. Find $\sum_{r=1}^{n}(r^3 - r)$ and factorise your answer. **[4]**

End of Worksheet • *Show all working clearly.*